Kaushal Raj

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Professional Summary

Final-year B.Tech Computer Science and Engineering student graduating in June 2027, with internship experience in machine learning and hands-on experience in backend development, REST APIs, and C++ programming. Developed a multi-tenant incident management platform using Node.js, Express.js, and MongoDB, along with modular C++ applications using STL.

Education

Asansol Engineering College — B.Tech, Computer Science and Engineering | Expected June 2027

Technical Skills

Programming Languages: C++, C, Java, Python, JavaScript
Frameworks & APIs: Node.js, Express.js, React.js, Flask, REST APIs
Databases: MySQL, MongoDB
Tools: Git, GitHub, Docker, Linux, VS Code, Postman, CMake
Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Experience

Machine Learning & Data Science Intern — Zaalima Private Limited
Aug 2026 – Present

- **Developed** Python-based data preprocessing and validation pipelines for machine learning and computer vision applications.
- **Processed and validated** annotated industrial defect datasets, including YOLO-format label preparation and data quality checks.
- **Implemented** NLP preprocessing workflows and a spaCy-based named entity recognition baseline for contract analysis.

Projects

SignalOps — Full-Stack Incident Management & Observability Platform
Tech Stack: React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Redis, BullMQ, Jest, Supertest

- **B**uilt REST APIs and a React dashboard to manage organizations, services, log ingestion, alerts, and incidents.
- **Implemented** JWT refresh-token rotation, organization-scoped access controls, and SHA-256-hashed service API keys for log ingestion.
- **Engineered** threshold-based alerts with deduplication, automatic incident creation, and append-only timelines; added an optional Redis/BullMQ worker.
- **C**reated backend tests with Jest, Supertest, and MongoDB Memory Server; configured GitHub Actions CI for testing.

C++ Chess Engine — Data Structures & Algorithms
Tech Stack: C++, STL, CMake, Git

- **D**esigned modular board representation and implemented piece encoding, square mapping, FEN parsing, and foundational move-generation logic.
- Developed material-based position evaluation and organized reusable components for testing and future search integration.

ChronoCLI — C++ Command-Line Calendar
Tech Stack: C++, STL, OOP, File Handling, CMake

- Built a command-line calendar supporting monthly views, weekday calculations, leap years, and input validation.
- Refactored calendar and event-management modules using OOP; implemented file-based event storage across sessions.

Achievements

- **Solved 300+ coding problems** across LeetCode and HackerRank.
- **Completed the 160 Days Coding Challenge** on GeeksforGeeks.

Certifications

- **Python for Data Science, AI & Development** — IBM (Coursera) | **Git & GitHub** — IBM (Coursera)
- **Open Source Software Development, Linux and Git** — Coursera | **Generative AI** — Internshala Trainings

Coding Profiles

- **LeetCode:** leetcode.com/u/rajkaushall/ | **GeeksforGeeks:** geeksforgeeks.org/user/quietsyntax/